cal ISR missions as well as preparing an armed fleet of Unmanned Combat Aerial Vehicles (UCAVs) and, following government policy, mostly procuring them from Indian manufacturers. "We were on a procurement spree. But the LoC incident exposed our vulnerability when we learnt that our bird was hacked by people operating across the border. It raises questions not only about cybersecurity but also supply chain issues in case of wartime," says a defence official. It's time to address these issues first before modernising the unarmed wing, he adds.

The discovery of the compromised drones forced the army to suspend multiple procurement orders. Among these was one for 200 medium-altitude logistics drones, intended for deployment along the LAC. Following the LoC incident, when the ministry of defence (MoD) demanded information about the quantum of Chinese components in military drones—a known threat—it opened an ill-disguised can of worms: several Indian drone-makers were found to rely heavily on Chinese parts.

Experts have long warned about cybersecurity vulnerabilities that Chinese components in Indian drones pose. The greatest danger lies in data leaks and unauthorised access to sensitive military operations. Backdoors—mechanisms that bypass security features—embedded in Chinese electronics, particularly within communication modules, cameras and control systems, could be exploited to extract intelligence and sensitive military information, disrupt military operations, hinder operational readiness or even seize control of drones mid-operation.

The Director General of Military Intelligence (DGMI) had in the past issued repeated directives barring the use of Chinese-origin components in sensitive security equipment. The directives mandated that hardware and software in such systems must not come from China. Manufacturers were also instructed to ensure rigorous testing to eliminate potential malware threats. This was hardly surprising, given that a law passed by Beijing in 2017 mandated that all Chinese tech firms share data with government agencies. In 2019, a cautious US banned the usage of drones and their components made in China.

During the military standoff with China along the LAC in Ladakh following the Galwan clash in June 2020, the restrictions were expanded to include equipment and spares from all countries sharing land borders with India. However, deliberate obfuscation by some domestic manufacturers and laxity in compliance measures have led to the current situation.

The threat, moreover, isn't limited to military drones; the burgeoning civilian drone market is equally at risk. Recognising the urgency of the threat, the ministry of electronics and information technology (MeitY) has now expanded its cybersecurity initiatives to include civilian drones.

"We have been moving too quickly on drone adoption, both military and civilian, without adequately addressing their vulnerabilities," admits a senior defence official. He reveals that there is still no mechanism in place to verify the origin of components in 'Make in India' drones.

The military is contemplating a

RAMAN RAINA



EAGLE ON HIS SHOULDER An Indian Army soldier accompanied by a drone on LoC patrol in Poonch, J&K



"WITH SUPPLY CHAINS HEAVILY RELIANT ON CHINESE COMPONENTS, THE FOCUS IS ON FINDING SOLUTIONS TO ELIMINATE THEIR PRESENCE. A COMPREHENSIVE FRAMEWORK WILL SOON BE DEVELOPED"

MAJOR GENERAL C.S. MANN
Additional Director General, Army Design Bureau